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VIA FOR SEMICONDUCTOR DEVICE AND METHOD

Abstract

A structure includes a first conductive feature in a first dielectric layer; a second dielectric layer over the first dielectric layer; and a second conductive feature extending through the second dielectric layer to physically contact the first conductive feature, wherein the second conductive feature includes a metal adhesion layer over and physically contacting the first conductive feature; a barrier layer extending along sidewalls of the second dielectric layer; and a conductive filling material extending over the metal adhesion layer and the barrier layer, wherein a portion of the conductive filling material extends between the barrier layer and the metal adhesion layer.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a divisional of U.S. patent application Ser. No. 18/358,803, filed on Jul. 25, 2023, which is a continuation of U.S. patent application Ser. No. 17/001,917, filed on Aug. 25, 2020, now U.S. Pat. No. 11,810,857 issued on Nov. 7, 2023, each application is incorporated herein by reference.

BACKGROUND

[0002] Integrated circuits include interconnect structures, which comprise metal lines and vias to serve as three-dimensional wiring structures. The function of the interconnect structures is to properly connect densely packed devices together.

[0003] Metal lines and vias are formed in the interconnect structure. Metal lines and vias are typically formed by damascene processes, in which trenches and via openings are formed in dielectric layers. A barrier layer is then deposited, followed by the filling of the trenches and via openings with copper. After a Chemical Mechanical Polish (CMP) process, the top surfaces of the metal lines are leveled, leaving metal lines and vias.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. **1** through **10** illustrate the cross-sectional views of intermediate stages in the formation of a via, in accordance with some embodiments.

[0006] FIGS. **11** through **16** illustrate the cross-sectional views of intermediate stages in the formation of a via, in accordance with some embodiments.

[0007] FIGS. **17A-C** and **18** illustrate the cross-sectional views of intermediate stages in the formation of various vias, in accordance with some embodiments.

[0008] FIG. **19** illustrates a flowchart for a method of forming a via, in accordance with some embodiments.

DETAILED DESCRIPTION

[0009] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0010] Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0011] A method of selectively forming an adhesion layer for a conductive feature is provided in accordance with various embodiments. The intermediate stages in the formation of the conductive feature are illustrated in accordance with some embodiments. Some variations of some embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. In accordance with some embodiments of the present disclosure, the formation of the adhesion layer includes depositing the material of the adhesion layer on an underlying conductive feature exposed by an opening. A barrier layer is formed on sidewalls of the opening and blocked from being formed on the adhesion layer by a sacrificial layer. The sacrificial layer resists adhesion of the barrier layer material such that the barrier layer is selectively grown on the sidewalls of the opening, with little or no barrier layer being formed on the sacrificial layer. After the barrier layer is formed, the sacrificial layer is removed. The opening is then filled with a metallic material such as copper, which is formed on the adhesion layer. In this manner, a conductive feature (e.g., a via) may be formed on another conductive feature (e.g., a metal line). By forming the adhesion layer on the underlying conductive feature, the thermal stability of the interface may be improved. Additionally, by blocking the formation of the barrier layer on the adhesion layer, the resistance of the interface may be reduced. Thus, the yield and the performance may be improved for a semiconductor device formed using the techniques described herein.

[0012] FIG. **1** illustrates a cross-sectional view of a package component **100**, in accordance with some embodiments of the present disclosure. The package component **100** may be a device wafer (such as a logic device wafer) including active devices such as transistors and/or diodes, and may include passive devices such as capacitors, inductors, resistors, or the like. In accordance with alternative embodiments of the present disclosure, the package component **100** is an interposer wafer, which may or may not include active devices and/or passive devices. In accordance with yet alternative embodiments of the present disclosure, the package component **100** is a package substrate strip, which may include package substrates with cores therein or core-less package substrates. In subsequent discussion, a device wafer is used as an example of the package component **100**. The teaching of the present disclosure may also be applied to interposer wafers, package substrates, packages, etc.

[0013] In accordance with some embodiments of the present disclosure, the package component **100** includes a semiconductor substrate **20** and features formed at a top surface of the semiconductor substrate **20**. The semiconductor substrate **20** may comprise crystalline silicon, crystalline germanium, silicon germanium, a III-V compound semiconductor such as GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, the like, or combinations thereof. The semiconductor substrate **20** may also be a bulk silicon substrate or a Silicon-On-Insulator (SOI) substrate, in some embodiments. Shallow Trench Isolation (STI) regions (not shown) may be formed in the semiconductor substrate **20** to isolate the active regions in the semiconductor substrate **20**. Although not shown, through-vias may be formed extending into the semiconductor substrate **20** to electrically interconnect features on opposite sides of the package component **100**.

[0014] In accordance with some embodiments of the present disclosure, the package component **100** is used to form a device die. In these embodiments, integrated circuit devices **22** are formed on a top surface of the semiconductor substrate **20**. Example integrated circuit devices **22** include Complementary Metal-Oxide Semiconductor (CMOS) transistors, resistors, capacitors, diodes, or

the like. The details of the integrated circuit devices **22** are not illustrated herein. In accordance with alternative embodiments, the package component **100** is used for forming interposers. In accordance with these embodiments, the substrate **20** may also be, for example, a dielectric substrate.

[0015] Further illustrated in FIG. **1** is a dielectric layer **24**. The dielectric layer **24** may be, for example, an Inter-Layer Dielectric (ILD) or an Inter-Metal Dielectric (IMD). In accordance with some embodiments of the present disclosure, the dielectric layer **24** is an ILD, in which contact plugs are formed. The corresponding dielectric layer **24** may be formed of Phospho Silicate Glass (PSG), Boro Silicate Glass (BSG), Boron-Doped Phospho Silicate Glass (BPSG), Fluorine-Doped Silicate Glass (FSG), a silicon oxide layer (formed using Tetra Ethyl Ortho Silicate (TEOS)), the like, or combinations thereof. The dielectric layer **24** may be formed using spin-on coating, Atomic Layer deposition (ALD), Flowable Chemical Vapor Deposition (FCVD), Chemical Vapor Deposition (CVD), Plasma-Enhanced Chemical Vapor Deposition (PECVD), Low-Pressure Chemical Vapor Deposition (LPCVD), or the like. In accordance with some embodiments of the present disclosure, the dielectric layer **24** is an IMD, in which metal lines and/or vias are formed. The corresponding dielectric layer **24** may be formed of a carbon-containing low-k dielectric material, Hydrogen SilsesQuioxane (HSQ), MethylSilsesQuioxane (MSQ), another low-k dielectric material, the like, or combinations thereof. In accordance with some embodiments of the present disclosure, the formation of the dielectric layer **24** includes depositing a porogen-containing dielectric material and then performing a curing process to drive out the porogen to form a dielectric layer **24** that is porous. The dielectric layer **24** may be formed on the integrated circuit devices **22**, or any number of dielectric layers and/or metallization layers may be present between the integrated circuit devices **22** and the dielectric layer **24**.

[0016] Still referring to FIG. **1**, a conductive feature **30** is formed in the dielectric layer **24**. The conductive feature **30** may be a metal line, a conductive via, a contact plug, or the like. In accordance with some embodiments, the conductive feature **30** includes a diffusion barrier layer **26** and a conductive filling material **28** over the diffusion barrier layer **26**. The diffusion barrier layer **26** may be formed of a conductive material such as titanium, titanium nitride, tantalum, tantalum nitride, the like, or combinations thereof. The conductive filling material **28** may be formed of copper, a copper alloy, aluminum, ruthenium, another metal or metal alloy, the like, or combinations thereof. The diffusion barrier layer **26** has the function of preventing the diffusion of the conductive material (such as copper) of the conductive filling material **28** from diffusing into the dielectric layer **24**. In accordance with some embodiments of the present disclosure, the formation of the conductive feature **30** may also adopt the techniques as discussed subsequently, such that the conductive feature **30** is formed on a metal adhesion layer, which may be similar to the metal adhesion layer **46** (see FIG. **6**), and/or such that the bottom portion of the diffusion barrier layer **26** is not formed.

[0017] As also shown in FIG. **1**, an etch stop layer **32** is formed over the dielectric layer **24** and the conductive feature **30**, in accordance with some embodiments. The etch stop layer **32** is formed of a dielectric material, which may include one or more materials such as aluminum oxide, aluminum nitride, a silicon oxide, a silicon nitride, silicon carbide, silicon oxynitride, silicon carbonitride, silicon oxycarbonitride, the like, or combinations thereof. The etch stop layer **32** may be formed of a material that has a high etching selectivity relative to the overlying dielectric layer **34** such that an etching of the dielectric layer **34** may stop on the etch stop layer **32**. In some embodiments, the etch stop layer **32** may have a thickness **T1** that is between about 2 nm and about 5 nm.

[0018] Still referring to FIG. **1**, a dielectric layer **34** is formed over the etch stop layer **32**. In some embodiments, the dielectric layer **34** is an IMD or an ILD. The dielectric layer **34** may comprise a dielectric material such as an oxide, a nitride, a carbon-containing dielectric material, the like, or combinations thereof. For example, the dielectric layer **34** may be formed of PSG, BSG, BPSG, FSG, TEOS oxide, HSQ, MSQ, the like, or combinations thereof. In some embodiments, the

dielectric layer **34** is a low-k dielectric layer having a dielectric constant value (k) that is lower than about 3.5 or that is lower than about 3.0.

[0019] FIGS. **2** through **10** illustrate a first process for forming a via **58** with a metal adhesion layer **46** (see FIG. **10**), in accordance with some embodiments. The examples shown in these and other figures show the formation of a via **58**, but the techniques described herein may be used to form other conductive features such as metal lines or the like, and all such conductive features are considered within the scope of this disclosure. It is also appreciated that the examples as shown in these figures describe a damascene process, but in other embodiments, a dual damascene process, in which a metal line, a via, a contact plug, or the like, is formed, is also contemplated. Additional dielectric layers may be formed and the process described in FIGS. **2-10** may be repeated to form additional layers of metallization (e.g., metal lines and vias). In this manner, the process described herein may be used to form an interconnect structure or the like.

[0020] In FIG. **2**, a via opening **42** is formed in the dielectric layer **34**, in accordance with some embodiments. The via opening **42** may be formed using, for example, suitable photolithography and etching techniques. In an example of a process of forming the via opening **42**, a metal hard mask **37** is first formed over the dielectric layer **34** and then patterned to form an opening **38**, as shown in FIG. **2**. The metal hard mask **37** may be formed of a material such as titanium nitride, boron nitride, another metal oxide or metal nitride, the like, or a combination thereof. The opening **38** in the metal hard mask **37** defines the pattern of a via (e.g., via opening **42** shown in FIG. **2**) that is subsequently filled to form a via (e.g., via **58** shown in FIG. **10**). In other embodiments, a photoresist structure may be formed over the dielectric layer **34** and patterned to form an opening that defines the pattern of a via.

[0021] The exposed dielectric layer **34** may then be etched to form an opening **42** extending into the dielectric layer **34** that exposes the etch stop layer **32**, as shown in FIG. **2**. The etching of the dielectric layer **34** may be performed using a wet etching process and/or a dry etching process (e.g., a plasma etching process). For example, the etching of the dielectric layer **34** may be performed using a process gas comprising fluorine and carbon, wherein fluorine is used for etching, with carbon having the effect of protecting the sidewalls of the resulting opening. With an appropriate fluorine and carbon ratio, the opening **42** may be formed having a desired profile. For example, the process gases for the etching may include one or more fluorine and carbon-containing gases such as C.sub.4F.sub.8, CH.sub.2F.sub.2, CF.sub.4, or the like, and may include one or more carrier gases such as Ar, N.sub.2, or the like. In an example of an etching process, the flow rate of C.sub.4F.sub.8 is in the range between about 0 sccm and about 50 sccm, the flow rate of CF.sub.4 is in the range between about 0 sccm and about 300 sccm (with at least one of C.sub.4F.sub.8 having a non-zero flow rate), and the flow rate of N.sub.2 is in the range between about 0 sccm and about 200 sccm. As an additional example, the process gases for the etching may include CH.sub.2F.sub.2 and a carrier gas such as N.sub.2. The flow rate of CH.sub.2F.sub.2 may be in the range between about 10 sccm and about 200 sccm, and the flow rate of N.sub.2 may be in the range between about 50 sccm and about 100 sccm. During the etching process, package component 100 may be kept at a temperature in the range between about 30° C. and about 60° C. In the etching process, plasma may be generated from the etching gases. The Radio Frequency (RF) power of the power source for the etching may be lower than about 700 Watts, and the pressure of the process gases is in the range from about 15 mTorr and about 30 mTorr. These are examples, and other etching processes or etching parameters are possible.

[0022] Next, referring to FIG. **3**, an etching process is performed to etch through the etch stop layer **32** and expose the conductive filling material **28**. In some cases, the etching process may also be referred to as a wet clean process when the etching of the etch stop layer **32** includes a wet etching process. In accordance with some embodiments of the present disclosure, the etching process uses a solution that includes glycol, dimethyl sulfide, amine, H.sub.2O.sub.2, the like, or combinations thereof. For example, glycol may be used as a surfactant, dimethyl sulfide may be used as a

solvent, amine may be used to remove undesirable organic residue on surfaces of the package component **100**, and/or a combination of H.sub.2O.sub.2 and amine may be used to etch the etch stop layer **32**.

[0023] Turning to FIG. **4**, a metal adhesion layer **46** is deposited on the exposed surface of the conductive filling material **28**, in accordance with some embodiments. As shown in FIG. **4**, the metal adhesion layer **46** may be deposited as a blanket layer or a conformal layer within the via opening **42** and on the metal hard mask **37** (when present). The metal adhesion layer **46** has the function of improving the adhesion between a first conductive feature (e.g., the conductive filling material **28**) and a second conductive feature (e.g., the conductive material **56** shown in FIG. **9**). For example, the metal adhesion layer **46** can improve the interface between conductive features when a barrier layer (e.g., the diffusion barrier layer **26** or the barrier layer **50** shown in FIG. **7**) is not present between the conductive features. In some cases, the use of the metal adhesion layer **46** can improve the thermal stability of the interface between the conductive features, which reduces the chance of the interface degrading or separating (e.g., “pull-up”). In this manner, the yield or the reliability of the connection between conductive features can be improved.

[0024] In some embodiments, the metal adhesion layer **46** may be a material such as cobalt, ruthenium, manganese, the like, or a combination thereof. For example, the metal adhesion layer **46** may be cobalt formed over a conductive filling material **28** of copper or may be manganese formed over a conductive filling material **28** of ruthenium, though other materials may be used for the metal adhesion layer **46** and/or the conductive filling material **28** than these examples. The metal adhesion layer **46** may be deposited using a suitable technique, such as physical vapor deposition (PVD) or the like.

[0025] In some embodiments, the metal adhesion layer **46** may be formed having a thickness T2 that is between about 20 Å and about 40 Å. The thickness T2 may be less than, about the same as, or greater than the thickness T1 of the etch stop layer **32**. In some embodiments, a directional deposition process like some PVD processes may form a metal adhesion layer **46** having a greater thickness on lateral surfaces than on sidewalls, as shown in FIG. **4**. For example, the metal adhesion layer **46** may be formed having a thickness T2 on lateral surfaces (e.g., on the conductive filling material **28**) and a thickness T3 on sidewalls that is about the same as or less than the thickness T2. In some embodiments, the thickness T3 of the metal adhesion layer **46** on sidewalls may be between about 5 Å and about 15 Å. Other thicknesses or combinations of thicknesses are possible.

[0026] Turning to FIG. **5**, an etching process is performed to thin the metal adhesion layer **46**. The etching process may reduce the thickness of the metal adhesion layer **46** on the conductive filling material **28**. For example, the etching process may reduce the thickness of the metal adhesion layer **46** on the conductive filling material **28** to a thickness T4 that is less than the deposited thickness T2. In some cases, after the etching process, the thickness T4 of the metal adhesion layer **46** on the conductive filling material **28** may be between about 5 Å and about 35 Å, although other thicknesses are possible. The thickness T4 may be less than, about the same as, or greater than the thickness T1 of the etch stop layer **32**. For example, FIG. **5** shows an embodiment in which the thickness of the metal adhesion layer **46** is less than the thickness of the etch stop layer **32**, but FIGS. **17A-C** shows embodiments in which some or all of the metal adhesion layer **46** has a thickness greater than the thickness of the etch stop layer **32**. After the etching process, the metal adhesion layer **46** on the conductive filling material **28** may have a flat surface, a convex surface, a concave surface, or an irregular surface, described in greater detail below for FIGS. **17A-C**. In some embodiments in which the metal adhesion layer **46** is deposited to a smaller thickness on sidewalls than on lateral surfaces, the etching process may remove the metal adhesion layer **46** from sidewalls of the via opening **42**, as shown in FIG. **5**.

[0027] The etching process to thin the metal adhesion layer **46** may include a suitable wet etching process or a suitable dry etching process. For example, the etching process may be an isotropic wet

etching process including H.sub.2O, HNO.sub.3, HCl, H.sub.2O.sub.2, the like, or a combination thereof. The wet etching process may also include other substances such as ethylenediamine tetra-acetic acid (EDTA), ethylene glycol tetra-acetic acid (EGTA), the like, or a combination thereof. For example, a wet etching process that may be used to thin a metal adhesion layer **46** of cobalt may include a mixture of H.sub.2O, HNO.sub.3, and EDTA. In some embodiments, the wet etching process includes a mixture of HNO.sub.3 at about 0.1 M and EDTA at between about 0.01 M and about 0.05 M, but other concentrations or mixtures are possible. The wet etching process may be performed at a temperature of between about 25° C. and about 40° C., and for a duration of time between about 10 seconds and about 60 seconds. These and other variations of an etching process are considered within the scope of this disclosure.

[0028] Turning to FIG. **6**, a sacrificial layer **48** is then formed on the metal adhesion layer **46**, in accordance with some embodiments. As shown in FIG. **6**, the sacrificial layer **48** may be deposited such that little or no material of the sacrificial layer **48** is formed on exposed surfaces of the dielectric layer **34**, described in greater detail below. The sacrificial layer **48** may be formed to block, prevent, or otherwise inhibit the subsequent formation of the barrier layer **50** (see FIG. **7**) on the metal adhesion layer **46**. In particular, the sacrificial layer **48** may block the barrier layer **50** from forming on the metal adhesion layer **46** over the conductive filling material **28**. By blocking formation of the barrier layer **50**, the resistance between the conductive filling material **28** and the subsequently deposited conductive material **56** (see FIG. **9**) may be reduced. In some embodiments, the sacrificial layer **48** may extend across the via opening **42** to completely cover the metal adhesion layer **46**. In some embodiments, the sacrificial layer **48** is formed having a thickness T5 that is between about 0.5 nm and about 3 nm. The top surface of the sacrificial layer **48** may be above, below, or about level with the etch stop layer **32**. In some cases, the sacrificial layer **48** may be formed on sidewall surfaces of the etch stop layer **32** within the via opening **42**, as shown in FIG. **6**.

[0029] In accordance with some embodiments, the sacrificial layer **48** comprises a material that adheres or bonds to the metal adhesion layer **46** and does not adhere or bond to the dielectric layer **34** and/or to the etch stop layer **32**. For example, the material may form chelation bonds with metal (e.g., cobalt) of the metal adhesion layer **46** but not form bonds with the dielectric layer **34**. In this manner, the subsequently formed barrier layer **50** (see FIG. **7**) may be formed on the dielectric layer **34** but be blocked from forming on the conductive filling material **28**. Additionally, the sacrificial layer **48** may be a material to which the subsequently formed barrier layer **50** is unlikely or unable to adhere or bond. For example, the chemical structure of the material may be hydrophobic and/or include non-polar groups to which precursors of the barrier layer **50** are unlikely or unable to bond, or the chemical structure of the material may inhibit adsorption of precursors of the barrier layer **50** due to steric hindrance. The material of the sacrificial layer **48** may be chosen such that one or more precursors of the subsequently formed barrier layer **50** have a high selectivity of adsorption on the dielectric layer **34** over the sacrificial layer **48**. For example, the selectivity of adsorption on the dielectric layer **34** versus adsorption on the sacrificial layer **48** may be greater than about 5:1. The selectivity may depend on the various materials and/or formation processes used. In this manner, the sacrificial layer **48** is not covered (or only partially covered) by the barrier layer **50** and thus the sacrificial layer **48** may be more easily removed. The sacrificial layer **48** may be deposited by a suitable technique such as a wet chemical soak or exposure to chemical gas, which may depend on the particular material(s) comprising the sacrificial layer **48**.

[0030] As a first example, the sacrificial layer **48** may include benzotriazole (BTA), which has the chemical formula C.sub.6H.sub.4N.sub.3H. BTA molecules have a first side with three nitrogen atoms that can bond to a metal such as cobalt and a second side which has a hydrophobic benzo ring to which some precursors of the barrier layer **50** are unable to bond. The first side of the BTA molecule can bond to the metal adhesion layer **46** while the second side protrudes and blocks

precursors from bonding to the metal adhesion layer **46**. In this manner, a sacrificial layer **48** comprising a monolayer of BTA or multiple monolayers of BTA can prevent the barrier layer **50** from forming on the metal adhesion layer **46** or on the sacrificial layer **48**. In some embodiments, a sacrificial layer **48** may be formed from BTA by soaking the package component **100** in a wet chemical solution containing BTA. For example, BTA may be part of a solution containing H_2O and/or H_2O_2 , though solutions having other compositions may be used. The solution may be heated to a temperature between about 25°C . and about 50°C ., and the package component **100** may be soaked for a duration of time between about 10 seconds and about 60 seconds. A wet clean process may be performed on the package component **100** after soaking in the solution. A sacrificial layer **48** including BTA may be formed using other solutions, process conditions, or techniques than these. The material and deposition technique described is an example, and the sacrificial layer **48** may be formed from other materials using a wet chemical soak process, such as thiol, phosphate, the like, other materials, or combinations thereof.

[0031] As a second example, the sacrificial layer **48** may include 5-Decyne, which has the chemical formula $\text{C}_{10}\text{H}_{18}$. The 5-Decyne molecules can form bonds to metals such as cobalt and to other 5-Decyne molecules, but 5-Decyne molecules do not bond to the dielectric layer **34**.

Additionally, precursors of the barrier layer **50** do not form bonds to 5-Decyne molecules. For example, 5-Decyne is hydrophobic with $-\text{CH}_3$ groups, and thus 5-Decyne does not bond to the barrier layer **50**. In this manner, a sacrificial layer **48** comprising a layer of 5-Decyne molecules can prevent the barrier layer **50** from forming on the metal adhesion layer **46** or on the sacrificial layer **48**. In some embodiments, a sacrificial layer **48** may be formed from 5-Decyne by exposing the package component **100** to a gas mixture including 5-Decyne molecules. For example, the 5-Decyne may be part of a gas mixture including carrier gases such as He, Ar, or the like, though other mixtures may be used. The gas mixture may be flowed into the process chamber at a flow rate between about 600 sccm and about 3000 sccm for a time duration between about 10 seconds and about 120 seconds. A process temperature between about 100°C . and about 350°C . may be used, and a process pressure between about 1 Torr and about 30 Torr may be used. A sacrificial layer **48** including 5-Decyne may be formed using other gas mixtures, process conditions, or techniques than these. The material and deposition technique described is an example, and the sacrificial layer **48** may be formed from other materials using a gas deposition process, such as from other alkynes, derivatives of alkenes, BTA in a gaseous phase, the like, hexane in a gaseous phase, the like, or a combination thereof.

[0032] Turning to FIG. 7, a conductive barrier layer **50** is deposited on surfaces within the via opening **42**, in accordance with some embodiments. The barrier layer **50** has the function of preventing atoms in the subsequently deposited conductive material **56** (FIG. 9) from diffusing into dielectric layer **34**. The barrier layer **50** is formed over exposed surfaces of the dielectric layer **34** and the etch stop layer **32**, but is blocked from forming on the metal adhesion layer **46** within the via opening **42** by the sacrificial layer **48**. Additionally, the barrier layer **50** does not significantly form on exposed surfaces of the sacrificial layer **48**, as described previously. The barrier layer **50** may comprise a barrier material such as titanium, titanium nitride, tantalum, tantalum nitride, the like, or a combination thereof. The barrier layer **50** may be a material similar to the diffusion barrier layer **26**, in some embodiments.

[0033] In some embodiments, the barrier layer **50** may be deposited using a suitable process, such as an ALD process and/or a CVD process. In some cases, forming the barrier layer **50** using an ALD process or a CVD process may allow for better step coverage and better conformity compared with other processes, such as a PVD process. For example, the barrier layer **50** may comprise TaN deposited using an ALD process. The precursors of TaN may include, for example, Pentakis Dimethylamino Tantalum ("PDMAT") as a first precursor, which has the chemical formula $\text{C}_{10}\text{H}_{30}\text{N}_5\text{Ta}$, and ammonia as a second precursor, which has the chemical formula NH_3 . The barrier layer **50** may include different materials and/or different precursors in other

embodiments. In some embodiments, the deposition of the barrier layer **50** may be performed in the same process chamber as the formation of the sacrificial layer **48** and/or the metal adhesion layer **46**. In some embodiments, the barrier layer **50** may be formed having a thickness **T6** that is between about 10 Å and about 40 Å.

[0034] Turning to FIG. **8**, the sacrificial layer **48** is removed, in accordance with some embodiments. In some embodiments, the process used to remove the sacrificial layer **48** also densifies the barrier layer **50**. The barrier layer **50** after densification is referred to herein as densified barrier layer **50'** or barrier layer **50'**, and is indicated similarly in FIG. **8** and subsequent figures. The removal process may densify the barrier layer **50** into densified barrier layer **50'** by, for example, driving nitrogen from the barrier layer **50**. A removal process that densifies the barrier layer **50** may include, for example, a thermal treatment or a plasma treatment such as an in-situ H.sub.2 or Ar plasma treatment, described in greater detail below. After removing the sacrificial layer **48**, the barrier layer **50'** may be separated from the metal adhesion layer **46** by a distance **T5'** that is approximately equal to the thickness **T5** of the sacrificial layer **48**. For example, the distance **T5'** may be between about 0.5 nm and about 3 nm, though other distances are possible. As shown in FIG. **8**, removing the sacrificial layer **48** may expose sidewalls of the etch stop layer **32** that were previously covered by the sacrificial layer **48**.

[0035] In some embodiments, a removal process that removes the sacrificial layer **48** includes a thermal treatment such as an anneal process. For example, the anneal process may include annealing the package component **100** in an anneal chamber at a temperature between about 250° C. and about 400° C. for a duration of time between about 30 seconds and about 300 seconds. The package component **100** may be exposed to one or more gases during the anneal process, such as an inert gas (e.g., He, Ar, or the like), a reducing gas (e.g., H.sub.2 or the like), or a combination thereof. The gas(es) may be flowed into the anneal chamber at a flow rate between about 600 sccm and about 3000 sccm. During the anneal process, the anneal chamber may have a pressure between about 1 Torr and about 30 Torr. A removal process including an anneal process may have other annealing parameters than these. In some embodiments, the anneal chamber is the same chamber as the process chamber used for depositing the barrier layer **50**.

[0036] In some embodiments, the removal process includes a plasma treatment. For example, the plasma treatment may include exposing the package component **100** to a plasma of one or more process gases such as H.sub.2, NH.sub.3, Ar, the like, or combinations thereof. The process gas(es) may be flowed at a flow rate between about 600 sccm and about 3000 sccm. The plasma treatment may be performed at a pressure between about 0.1 Torr and about 5 Torr. In some embodiments, the plasma is generated using an RF power between about 100 Watts and about 600 Watts. The plasma treatment may be performed at a temperature between about 25° C. and about 400° C., and may be performed for a duration of time between about 10 seconds and about 30 seconds. A removal process including a plasma treatment may have other parameters than these. In some embodiments, the plasma treatment is performed using the same process chamber used for depositing the barrier layer **50**. In some embodiments, one of an anneal process or a plasma treatment is performed. In other embodiments both an anneal process and a plasma treatment are performed, which may be performed in either order.

[0037] In some cases, a removal process including an thermal treatment and/or a plasma treatment as described above may reduce the concentration of nitrogen within the barrier layer **50**, forming a densified barrier layer **50'**. A densified barrier layer **50'** may be more effective at blocking diffusion into the dielectric layer **34** and thus may improve device performance. In some cases, the removal process may also reduce the resistivity of the barrier layer **50**, which can further improve device performance. While the figures illustrate embodiments including a densified barrier layer **50'**, in other embodiments a removal process may be performed that does not densify the barrier layer **50**.

[0038] Turning to FIG. **9**, a conductive material **56** is deposited to fill the via opening **42**, in accordance with some embodiments. The conductive material **56** may be deposited, for example,

by performing a blanket deposition to form a metal seed layer (e.g., a copper layer) using a PVD process, and then filling the rest of via opening **42** using, for example, electro-plating, electro-less plating, deposition, or the like. The conductive material **56** may include copper, a copper alloy, cobalt, tungsten, ruthenium, the like, other metals, or combinations thereof. The conductive material **56** may be the same material as the conductive filling material **28** or may be a different material.

[0039] By blocking the barrier layer **50** from forming over the conductive filling material **28** (see FIG. 7), a “barrier-free” interface between the conductive material **56** and the conductive filling material **28** is formed. In some cases, the contact resistance (R_c) of this barrier-free interface is smaller than if the barrier layer **50** was present between the conductive material **56** and the conductive filling material **28**. In some cases, the use of the metal adhesion layer **46** without an overlying barrier layer **50** may reduce the contact resistance between about 20% and about 40%. Additionally, by forming the metal adhesion layer **46** on the conductive filling material **28**, the thermal stability of the interface between the conductive material **56** and the conductive filling material **28** may be improved. In this manner, the techniques described herein may allow for improved contact resistance between a via **58** and a conductive feature (e.g., conductive feature **30**). As shown in FIG. 9, due to the separation between the barrier layer **50'** and the metal adhesion layer **46**, the conductive material **56** may contact sidewalls of the etch stop layer **32** that are not covered by the barrier layer **50'**. In this manner, portions of the conductive material **56** may extend between the barrier layer **50'** and the metal adhesion layer **46**, and the interface between the conductive material **56** and the conductive filling material **28** may be formed having a greater lateral width. In some cases, increasing the lateral width of the interface can decrease the resistance of the interface.

[0040] Turning to FIG. 10, a planarization process such as a Chemical Mechanical Planarization (CMP) process, a mechanical polish process, and/or a grinding process may be performed to remove excess portions of conductive material **56**, hence forming the via **58**, in accordance with some embodiments. The via **58** includes remaining portions of the barrier layer **50'** and the conductive material **56**. FIG. 10 also illustrates the formation of an optional dielectric etch stop layer **62**, which covers and contacts the dielectric layer **34** and the via **58**. In accordance with some embodiments, the dielectric etch stop layer **62** is formed of one or more layers of metal oxide, metal nitride, metal carbonitride, silicon nitride, the like, or combinations thereof. In this manner, a via **58** may be formed over a conductive feature **30** using a metal adhesion layer **46**.

[0041] FIGS. 11 through 16 illustrate a second process for forming a via **58** with a metal adhesion layer **46** (see FIG. 16), in accordance with some embodiments. The process shown in FIGS. 11 through 16 is similar to the first process shown in FIGS. 1 through 10, except that the metal adhesion layer **46** is formed using a selective deposition process that forms the material of the metal adhesion layer **46** on surfaces of the conductive filling material **28** over surfaces of the dielectric layer **34**. FIG. 11 follows from the structure shown in FIG. 3, in which the conductive filling material **28** is exposed. The process described in FIGS. 11-16 may be repeated to form additional layers of metallization, such as for forming an interconnect structure or the like.

[0042] FIG. 11 illustrates the deposition of the metal adhesion layer **46**, in accordance with some embodiments. The metal adhesion layer **46** may be a similar material as those described previously for FIG. 4. In the embodiment shown in FIG. 11, the metal adhesion layer **46** is deposited using a selective deposition process such that the material of the metal adhesion layer **46** is formed on the conductive filling material **28** and little or no material of the metal adhesion layer **46** is formed on the dielectric layer **34**. For example, the selectivity of deposition on the conductive filling material **28** versus the dielectric layer **34** may be greater than about 100:1, though other selectivities are possible. In some cases, the material of the metal adhesion layer **46** may also be formed on the metal hard mask **37** (if present).

[0043] In some embodiments, the metal adhesion layer **46** may be deposited on the conductive

filling material **28** to a thickness $T7$ that is between about 10 Å and about 40 Å, although other thicknesses are possible. The thickness $T7$ may be less than, about the same as, or greater than the thickness $T1$ of the etch stop layer **32**. For example, FIG. **11** shows an embodiment in which the thickness of the metal adhesion layer **46** is less than the thickness of the etch stop layer **32**, but FIGS. **17A-C** shows embodiments in which some or all of the metal adhesion layer **46** has a thickness greater than the thickness of the etch stop layer **32**. The metal adhesion layer **46** on the conductive filling material **28** may have a flat surface, a convex surface, a concave surface, or an irregular surface, described in greater detail below for FIGS. **17A-C**.

[0044] To selectively deposit the metal adhesion layer **46**, a suitable deposition process may be used, such as ALD, PVD, CVD, or the like. The particular deposition process used and/or precursors used may depend on the specific materials of the metal adhesion layer **46**, the conductive filling material **28**, and/or the dielectric layer **34**. As an example of selectively depositing the metal adhesion layer **46** on the conductive filling material **28**, a CVD process may be used to selectively deposit a metal adhesion layer **46** of cobalt. The cobalt may be deposited by a CVD process using cyclopentadienylcobalt dicarbonyl ($\text{CpCo}(\text{CO})_{\text{sub.2}}$) and another gas such as $\text{H}_{\text{sub.2}}$ or a mixture of $\text{H}_{\text{sub.2}}$ and $\text{NH}_{\text{sub.3}}$. A process temperature between about 150° C. and about 250° C. may be used, and a process pressure between about 10 Torr and about 30 Torr may be used. In some cases, cobalt deposited in this manner may deposit on the dielectric layer **34** to a thickness that is less than 1% of the thickness of the cobalt that is deposited on the conductive filling material **28**. As another example, a metal adhesion layer **46** of ruthenium may be selectively deposited using “CHORuS” ($\text{Ru}(\text{C}_{\text{sub.7}}\text{H}_{\text{sub.10}})(\text{CO})_{\text{sub.3}}$) or triruthium dodecacarbonyl ($\text{Ru}_{\text{sub.3}}(\text{CO})_{\text{sub.12}}$). In some embodiments, a surface treatment may be performed on the conductive filling material **28** prior to deposition of the metal adhesion layer **46**. The surface treatment may, for example, remove oxidation or otherwise clean the surface of the conductive filling material **28**, and may include a wet chemical treatment and/or a plasma treatment, such as exposing the surface of the conductive filling material **28** to a hydrogen plasma or the like. These are examples, and other materials, precursors, gases, and/or process parameters may be used in other embodiments.

[0045] Turning to FIG. **12**, a sacrificial layer **48** is then formed on the metal adhesion layer **46**, in accordance with some embodiments. The sacrificial layer **48** may be similar to the sacrificial layer **48** described for FIG. **6**, and may be formed in a similar manner. For example, the sacrificial layer **48** may be selectively deposited on the metal adhesion layer **46**. The sacrificial layer **48** may be formed to block, prevent, or otherwise inhibit the subsequent formation of the barrier layer **50** (see FIG. **13**) on the metal adhesion layer **46**. In some embodiments, the sacrificial layer **48** is formed having a thickness $T8$ that is between about 0.5 nm and about 3 nm. The top surface of the sacrificial layer **48** may be above, below, or about level with the etch stop layer **32**. In some cases, the sacrificial layer **48** may be formed on sidewall surfaces of the etch stop layer **32** within the via opening **42**, as shown in FIG. **12**.

[0046] Turning to FIG. **13**, a conductive barrier layer **50** is deposited on surfaces within the via opening **42**, in accordance with some embodiments. The barrier layer **50** may be similar to the barrier layer **50** described for FIG. **7**, and may be formed in a similar manner. The barrier layer **50** is formed over exposed surfaces of the dielectric layer **34** and the etch stop layer **32**, but is blocked from forming on the metal adhesion layer **46** within the via opening **42** by the sacrificial layer **48**. Additionally, the barrier layer **50** does not significantly form on exposed surfaces of the sacrificial layer **48**, as described previously. The barrier layer **50** may also be formed over the metal hard mask **37** (if present), as shown in FIG. **13**. In some embodiments, the deposition of the barrier layer **50** may be performed in the same process chamber as the formation of the sacrificial layer **48** and/or the metal adhesion layer **46**. In some embodiments, the barrier layer **50** may be formed having a thickness $T9$ that is between about 10 Å and about 40 Å.

[0047] Turning to FIG. **14**, the sacrificial layer **48** is removed, in accordance with some

embodiments. The sacrificial layer **48** may be removed using a technique similar to those described previously for FIG. **8**. The removal process may densify the barrier layer **50** into densified barrier layer **50'** by, for example, driving nitrogen from the barrier layer **50** using a thermal treatment or a plasma treatment similar to that described previously. In other embodiments, the removal process does not densify the barrier layer **50**. After removing the sacrificial layer **48**, the barrier layer **50'** may be separated from the metal adhesion layer **46** by a distance $T8'$ that is approximately equal to the thickness $T8$ of the sacrificial layer **48**. For example, the distance $T8'$ may be between about 0.5 nm and about 3 nm, though other distances are possible. As shown in FIG. **8**, removing the sacrificial layer **48** may expose sidewalls of the etch stop layer **32** that were previously covered by the sacrificial layer **48**.

[0048] Turning to FIG. **15**, a conductive material **56** is deposited to fill the via opening **42**, in accordance with some embodiments. The conductive material **56** may be similar to the conductive material **56** described for FIG. **9**, and may be formed in a similar manner. In FIG. **16**, a planarization process may be performed to remove excess portions of conductive material **56**, hence forming the via **58**, in accordance with some embodiments. FIG. **16** also illustrates the formation of an optional dielectric etch stop layer **62**, which may be similar to the dielectric etch stop layer **62** described for FIG. **10**. In this manner, a via **58** may be formed over a conductive feature **30** using a metal adhesion layer **46** formed by a selective deposition process.

[0049] Turning to FIGS. **17A-C**, metal adhesion layers **46** having different top surfaces are shown, in accordance with some embodiments. FIGS. **17A-C** illustrate magnified portions of cross-sections similar to the cross-sections shown previously (e.g., FIG. **10** or FIG. **16**). Some or all of the metal adhesion layers **46** shown in FIGS. **17A-C** have a thickness greater than the thickness $T1$ of the etch stop layer **32**, but in other embodiments some or all of a metal adhesion layer **46** may have a thickness about the same as or less than the thickness $T1$ of the etch stop layer **32**. FIG. **17A** illustrates an embodiment in which the metal adhesion layer **46** has a substantially flat top surface. A metal adhesion layer **46** with a flat surface may be formed, for example, by conformal deposition followed by thinning (e.g., as described for FIGS. **4-5**) or by selective deposition (e.g., as described for FIG. **11**). FIG. **17B** illustrates an embodiment in which the metal adhesion layer **46** has a substantially convex top surface. A metal adhesion layer **46** with a convex surface may be formed, for example, by selective deposition (e.g., as described for FIG. **11**). In some cases, the convex shape may be formed due to the smaller deposition rate of the material of the metal adhesion layer **46** on the dielectric layer **34**. FIG. **17C** illustrates an embodiment in which the metal adhesion layer **46** has a substantially concave top surface. A metal adhesion layer **46** with a concave surface may be formed, for example, by conformal deposition followed by thinning (e.g., as described for FIGS. **4-5**). In some cases, the concave shape may be formed due to the etching process that thins the metal adhesion layer **46** removing less material near surfaces of the dielectric layer **34**. (e.g., near the sidewalls of the via opening **42**.) In some cases, a metal adhesion layer **46** with a concave surface may have a smaller electrical resistance than a metal adhesion layer **46** with a convex surface. The shapes of the metal adhesion layer **46** shown in FIGS. **17A-C** are examples, and a metal adhesion layer **46** may be formed having a differently shaped top surface, such as an irregular shape or a combination of flat, convex, or concave surfaces.

[0050] FIG. **18** illustrates an embodiment in which an additional metal adhesion layer **60** is deposited before the deposition of the conductive material **56**. The additional metal adhesion layer **60** may be a material such as cobalt, ruthenium, the like, or combinations thereof. The additional metal adhesion layer **60** may be the same material or a different material than the underlying metal adhesion layer **46**. The additional metal adhesion layer **60** may be formed using a suitable deposition process such as PVD, CVD, ALD, or the like, including the techniques described above for deposition the metal adhesion layer **46**. In some embodiments, the additional metal adhesion layer **60** may be formed having a thickness $T10$ that is between about 10 Å and about 40 Å. In some cases, depositing an additional metal adhesion layer **60** may improve the adhesion and

thermal stability of the interface between the conductive material **56** and the barrier layer **50'**.

[0051] FIG. **19** illustrates a flowchart for a method **200** of forming a via, in accordance with some embodiments. It should be understood that the embodiment method shown in FIG. **19** is merely an example of many possible embodiment methods. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. **19** may be added, removed, replaced, rearranged and repeated.

[0052] Referring to FIG. **19**, at step **201**, an opening is etched through a dielectric layer to expose a conductive feature. This is shown, for example, in FIG. **3**. At step **202**, a metal adhesion layer is deposited on the conductive feature in the opening. This is shown, for example, in FIG. **4** in which a metal adhesion layer is conformally deposited or in FIG. **11** in which a metal adhesion layer is selectively deposited. At step **203**, a sacrificial layer is deposited on the metal adhesion layer in the opening. This is shown, for example, in FIG. **6** or FIG. **12**. At step **204**, a barrier layer is deposited in the opening. This is shown, for example, in FIG. **7** or FIG. **13**. At step **205**, the sacrificial layer is removed. This is shown, for example, in FIG. **8** or FIG. **14**. In some embodiments, removing the sacrificial layer also densifies the barrier layer. At step **206**, a conductive material is deposited on the metal adhesion layer in the opening, forming the via. This is shown, for example, in FIG. **10** or FIG. **16**.

[0053] The embodiments of the present disclosure have some advantageous features. By using a sacrificial layer to block formation of the barrier layer on a conductive feature, the contact resistance of the interface between the conductive feature and an overlying conductive material may be reduced, which can improve device performance. Additionally, the formation of a metal adhesion layer between the conductive feature and the overlying conductive material can improve the thermal stability of the interface, which can reduce time-dependent dielectric breakdown (TDDB) of the device and improve yield. The metal adhesion layer can be formed using different processes, such as by a conformal deposition process followed by a thinning process or by a selective deposition process that selectively deposits the metal adhesion layer on the conductive feature. The techniques described herein may be used to form a variety of conductive features, such as metal lines, vias, interconnects, Back End of Line (BEOL) features, or the like.

[0054] In accordance with some embodiments of the present disclosure, a structure includes a first conductive feature in a first dielectric layer; a second dielectric layer over the first dielectric layer; and a second conductive feature extending through the second dielectric layer to physically contact the first conductive feature, wherein the second conductive feature includes a metal adhesion layer over and physically contacting the first conductive feature; a barrier layer extending along sidewalls of the second dielectric layer; and a conductive filling material extending over the metal adhesion layer and the barrier layer, wherein a portion of the conductive filling material extends between the barrier layer and the metal adhesion layer. In an embodiment, the structure includes an etch stop layer between the first dielectric layer and the second dielectric layer, wherein the second conductive feature extends through the etch stop layer. In an embodiment, a thickness of the metal adhesion layer is less than a thickness of the etch stop layer. In an embodiment, the etch stop layer is free of the barrier layer. In an embodiment, the barrier layer includes tantalum nitride (TaN). In an embodiment, a top surface of the metal adhesion layer is concave. In an embodiment, a top surface of the metal adhesion layer is convex. In an embodiment, the portion of the conductive filling material extending between the barrier layer and the metal adhesion layer has a thickness between 0.5 nm and about 3 nm.

[0055] In accordance with some embodiments of the present disclosure, an integrated circuit structure includes a first inter-metal dielectric layer (IMD) over a substrate; a metal line in the first IMD; a second IMD over the first IMD; and a via in the second IMD, wherein the via physically contacts the metal line, wherein the via includes a filling material including a first conductive material; a first adhesion layer extending between the filling material and the metal line, wherein the first adhesion layer includes a second conductive material; and a diffusion barrier layer

extending between the filling material and the second IMD, wherein the first adhesion layer is free of the diffusion barrier layer, wherein the diffusion barrier layer includes a third conductive material. In an embodiment, the first conductive material is copper and the second conductive material is cobalt. In an embodiment, the first adhesion layer has a thickness between 5 Å and 35 Å. In an embodiment, the integrated circuit structure includes an etch stop layer extending over the metal line, wherein the via extends through the etch stop layer. In an embodiment, a sidewall of the etch stop layer is covered by the first adhesion layer. In an embodiment, the filling material physically contacts a sidewall of the etch stop layer. In an embodiment, the integrated circuit structure includes a second adhesion layer extending over the diffusion barrier layer and the first adhesion layer. In an embodiment, the second adhesion layer includes the second conductive material.

[0056] In accordance with some embodiments of the present disclosure, a method of forming a semiconductor device includes forming a conductive feature in a first dielectric layer; forming a second dielectric layer over the first dielectric layer; etching an opening through the second dielectric layer, the etching exposing a surface of the conductive feature; depositing a metal adhesion layer in the opening, wherein the metal adhesion layer covers the exposed surface of the conductive feature; depositing a sacrificial layer in the opening, wherein the sacrificial layer selectively forms on the metal adhesion layer over surfaces of the second dielectric layer; depositing a barrier layer in the opening, wherein the barrier layer selectively forms on surfaces of the second dielectric layer over the sacrificial layer; removing the sacrificial layer; and depositing a conductive material to fill the opening, wherein the conductive material covers the metal adhesion layer. In an embodiment, the sacrificial layer includes benzotriazole (BTA). In an embodiment, depositing the metal adhesion layer includes a chemical vapor deposition (CVD) process that selectively forms the metal adhesion layer on the conductive feature over the second dielectric layer. In an embodiment, depositing the metal adhesion layer includes conformally depositing the metal adhesion layer within the opening and, after depositing the metal adhesion layer, performing an etching process to thin the metal adhesion layer.

[0057] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: forming an insulating layer over a conductive feature; forming a metal mask over the insulating layer; etching the insulating layer using the metal mask to form an opening that exposes a surface of the conductive feature and sidewalls of the insulating layer; depositing an adhesion layer as a blanket layer that covers the exposed surface of the conductive feature, the sidewalls of the insulating layer, and the metal mask; performing an etching process that removes the adhesion layer from the sidewalls of the insulating layer, wherein the surface of the conductive feature remains covered by the adhesion layer after the etching process; selectively depositing a sacrificial layer on the adhesion layer; depositing a barrier layer on surfaces of the insulating layer and the metal mask; performing a removal process that removes the sacrificial layer and densifies the barrier layer; and depositing a conductive material in the opening.
2. The method of claim 1, wherein the adhesion layer comprises at least one of cobalt, ruthenium, or manganese.

3. The method of claim 1 further comprising, after depositing the conductive material, removing the metal mask.
 4. The method of claim 1 further comprising performing a planarization process, wherein top surfaces of the barrier layer and the conductive material are level after the planarization process.
 5. The method of claim 1, wherein the barrier layer and the adhesion layer are physically separated by the conductive material.
 6. The method of claim 1, wherein the sacrificial layer comprises 5-Decyne.
 7. The method of claim 1, wherein the removal process comprises a plasma process.
 8. The method of claim 1, wherein the removal process removes nitrogen from the barrier layer.
 9. A method comprising: forming a conductive feature in a first dielectric layer; depositing a second dielectric layer over the first dielectric layer; forming a patterned mask on the second dielectric layer; using the patterned mask, etching an opening in the second dielectric layer that exposes a surface of the conductive feature and a surface of the second dielectric layer; depositing an adhesion layer on the patterned mask and on the exposed surface of the conductive feature; depositing a sacrificial layer on the adhesion layer, wherein the exposed surface of the second dielectric layer is free of the sacrificial layer; depositing a barrier layer on the exposed surface of the second dielectric layer; performing a plasma treatment process on the barrier layer; and depositing a conductive material on the adhesion layer.
 10. The method of claim 9, wherein the adhesion layer is a continuous layer.
 11. The method of claim 9 further comprising thinning the adhesion layer using an etching process.
 12. The method of claim 11, wherein the etching process removes the adhesion layer from the exposed surface of the second dielectric layer.
 13. The method of claim 9, wherein the plasma treatment process removes the sacrificial layer.
 14. The method of claim 9, wherein the plasma treatment process increases a density of the barrier layer.
 15. The method of claim 9, wherein the sacrificial layer is deposited on a surface of patterned mask.
 16. A method comprising: depositing an etch stop layer over a metal line; depositing a dielectric layer over the etch stop layer; forming an etch mask on the dielectric layer; using the etch mask, forming an opening through the etch stop layer and the dielectric layer, wherein the opening exposes the metal line; depositing a first adhesion material on the metal line and on the etch mask; depositing a hydrophobic material on the first adhesion material; depositing a metal nitride material in the opening, wherein the metal nitride material selectively bonds to the dielectric layer more than the metal nitride material bonds to the hydrophobic material; removing the hydrophobic material; and depositing a conductive material on the first adhesion material and on the metal nitride material.
 17. The method of claim 16 further comprising, before depositing the hydrophobic material, etching the first adhesion material.
 18. The method of claim 16 further comprising removing the etch mask after removing the hydrophobic material.
 19. The method of claim 16, wherein removing the hydrophobic material comprises performing an anneal process.
 20. The method of claim 16, wherein the conductive material directly contacts the etch stop layer.
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